Ornstein–Uhlenbeck Basin Segmentation of Gaze Trajectories

1 Some background to the OU framework

(See Appendix for a full derivation.)

The Fokker–Planck equation describes the change in probability density over time for an ensemble in $\mathbb{R}^d$. Divergence notation:

$$\partial_t p = -\nabla \cdot (fp) + \nabla^2 (Dp)$$

or in sum notation:

$$\partial_t p = -\sum_{i=1}^{d} \partial_i (f_i p) + \sum_{i=1}^{d} \partial_i \partial_i (Dp)$$

where f is the drift function, D is the diffusion matrix, and p is the probability density function (pdf).

The probability flux is defined as

$$J = fp - D\nabla p, \qquad \nabla \cdot J = \nabla \cdot (fp) - D\nabla^2 p,$$

so that $\nabla \cdot J = -\partial_t p$.

Where the diffusion coefficient is empirically derived:

$$D = \frac{\sigma^2}{2} I, \qquad \sigma^2 = \frac{1}{T-1} \sum_{k=1}^{T-1} \frac{\|S_{k+1} - S_k\|^2}{2\,\Delta t_{k+1}}.$$

The OU model assumes the following:

$$\nabla \cdot J = 0, \qquad J = 0;$$

the probability flux and net flow are exactly 0. More concretely, the pull of any region is inherent, any particular region's pull does not change in time or due to memory of itself or other regions, regardless of face identity.

By rewriting $f = -\nabla U$ we can write f in terms of p, which is standard for a Boltzmann distribution, that translates the process into the Ornstein–Uhlenbeck (OU) model

$$fp = D\nabla p \;\rightarrow\; f = D\nabla \log p \qquad \text{since } J = 0,$$

$$\nabla^\top (f) = \Lambda_k = -D\big[\nabla\nabla^\top \log \hat{p}\,\big]_{\mu_k}$$

is used to determine the OU mean reverters / basins.

Note that the model is meant to segment the trajectory along the image with regards to discovered AOIs, but it is a poor predictor of the actual dynamics.

2 Distributions over each image as a measurable space

The eye tracker data records the position $S_k = (x_k, y_k)$ for $k = 1, \ldots, T$, where $S_k \in \Omega_T$, $\Omega_T \subset \mathbb{R}^2$;

$$\Omega_T = [0, L_x] \times [0, L_y]$$

is a coordinate grid of real numbers, and T is the total number of samples.

The evaluation grid G

1. We choose some grid width h_g to input into the kernel density sum, which also constrains the KDE width.
2. Since corner boundaries are noisy, we partition with a half shift in the nodes:

$$d_i = \frac{L_i}{h_g}, \quad \text{where } h_g \text{ is divisible for all } L_i,$$

$$g_{i1} = \frac{h_g}{2} \;\to\; g_{id_i} = g_{i(d_i-1)} + h_g \;\to\; g_{id_i} = h_g\big(d_i - \tfrac{1}{2}\big) = L_i - \tfrac{h_g}{2},$$

so G_i runs from $\frac{h_g}{2}$ to $L_i - \frac{h_g}{2}$.

h_g is also a lower bound for the KDE width h we can choose, since the density would otherwise spill out.
We use a 2D Gaussian kernel for the weights:

$$\hat{p}(x,y) = \frac{1}{2\pi h^2 T} \sum_{k=1}^{T} \exp\left(-\frac{\|x-x_k\|^2 + \|y-y_k\|^2}{2h^2}\right), \qquad x, y \in G.$$

We then use the estimator to create candidates independent of pre-labelled regions.

Conditions for obtaining the basins (AOI) over the KDE

Define the neighborhood over the image for a single 2D neighbor with numpy convention

$$\mathcal{N}(x,y) = \{(y+a,\, x+b) : a, b \in \{-h_g, 0, h_g\}\} \quad \forall\, x, y \in G.$$

1. Set a density floor, i.e. pick a threshold of contribution c (bottom 60% contribution); $1-c$ are kept (top 40% contribution).
2. $$\hat{p}_{yx} \geq \max_{(y',x') \in \mathcal{N}(y,x)} \hat{p}_{y'x'} \implies (x,y) = \mu_k.$$
3. Compute $\det|\Lambda_k - I\lambda_i| = 0$, where $\lambda_{\min}(\Lambda_k) = \min(\lambda_i \,|\, \Lambda_k)$,

$$B(\Lambda_k) = \mathbf{1}\{\lambda_{\min}(\Lambda_k) > 0\};$$

Λ_k survives if $B(\Lambda_k) = 1$.

3 Pipeline and fitting procedure

The partition is produced in three steps, each reading the previous one's output: extraction of samples and trial structure from the eye-tracker recordings, face regions from the stimulus images, and estimation of $\hat{p}$, the basins and the visits per participant and image. Validation (Section 4) and results (Section 5) read only these outputs. Implementation details, parameters and usage are in the accompanying README.

3.1 Data

29 sessions from 18 participants, recorded at 500 Hz. The D cohort was recorded binocularly and the T cohort from the left eye. Each session has 122 trials, the first two practice; each image is shown once, for 6 s. Of the 120 images per session, 60 are IAPS pictures shared by all participants and 60 are ABM pictures specific to the participant; within each set, 20 are Negative, 20 Positive and 20 Neutral. Stimuli are 1024×768 px, centred on the 1280×1024 screen, so screen coordinates equal image coordinates plus $(128, 128)$.

3.2 Extraction

Each recording is parsed once. Trial boundaries, crosshair and stimulus onsets come from the message lines, with each display message corrected by its reported display delay. Trial variables (image and condition) are logged after the end of the trial and are assigned back to it. Every sample is assigned its trial and phase; the stimulus phase runs from stimulus onset to stimulus offset. EyeLink's own fixation and saccade events are kept. Their eye label is not filtered, so a binocular recording contributes both eyes' saccades while the sample coordinates are the left eye's.

3.3 Sample cleaning and pooling

Only stimulus-phase samples of non-practice trials are used. A sample is removed if it is missing or off-screen, if it lies within 80 ms of a missing sample, if it lies within an EyeLink saccade extended by 4 ms on each side, or if its speed exceeds 1000°/s at 44 px per degree. Trials keeping fewer than 20 samples are dropped. Because each image is shown once, the samples of one participant on one image form one estimation unit; images with fewer than 200 retained samples are dropped. T_{im} is the retained count for participant i and image m.

3.4 Face regions

Faces are detected in every stimulus image and fits 68 landmarks to each. The face ellipse is the second-moment fit of the landmarks: centre at their mean c_v, axes along the eigenvectors of their covariance, semi-axes a_v, b_v at two standard deviations scaled by 1.06, and tilt ϕ_v. With (u, v) the displacement from c_v rotated by ϕ_v,

$$Q_v(x) = \left(\frac{u}{a_v}\right)^2 + \left(\frac{v}{b_v}\right)^2,$$

inside the face when $Q_v \le 1$. The centre is the landmark centroid, near the nose. All detections are used. The ellipses are computed from the images alone and are never used in estimating $\hat{p}$ or the basins.

3.5 Estimation per image

Section 2 is carried out for each participant and image, for one bandwidth h.

Diffusion. σ^2 is the mean of $\|S_{k+1} - S_k\|^2/2\Delta t$ over adjacent retained samples with $\Delta t \le 4$ ms, and $D = \sigma^2/2$.

Density and basins. $\hat{p}$ is evaluated on G with $h_g = 16$ px; $\log \hat{p}$ is floored at 10^{-7} and differentiated by finite differences. The three basin conditions of Section 2 select the centres μ_k; the Hessian is symmetrised before its eigenvalues are taken. θ_1, θ_2 are the eigenvalues of Λ_k and θ their mean.

Membership. A sample x belongs to the basin minimising

$$Q_k(x) = \frac{(x-\mu_k)^\top \Lambda_k (x-\mu_k)}{\sigma^2} = \tfrac{1}{2}(x-\mu_k)^\top \big[-\nabla\nabla^\top \log \hat{p} \big]_{\mu_k} (x-\mu_k)$$

if $\min_k Q_k \le 1$, and is otherwise unlabelled. D cancels, so the partition depends on $\hat{p}$ alone.

3.6 Visits

A visit is a maximal run of consecutive samples with the same label within a trial; a run is broken by a label change or by a gap longer than 60 ms, and kept if it lasts at least 8 ms. Dwell Δ_{dwell} is the visit's duration. The re-entry index r counts earlier visits to the same basin in the same image ($r = 0$ is the first). A visit is *truncated* if it ends within 20 ms of stimulus offset. A basin is on a face if $\min_v Q_v(\mu_k) \le 1$; the minimising face is its face identity and $q = \min_v Q_v(\mu_k)$ its depth, with $q = 0$ at the face centre and $q = 1$ on the boundary.

4 Validation

4.1 Bandwidth selection (h)

The model treats the images independently, which avoids any bias in mapping out the landscape of regions. The bandwidth should show consistent results relative to the face labelling done independently.

A convenient measure of the bandwidth over the empirical data is the Silverman criterion in 2D

$$\hat{h} = \hat{\sigma}\left(\frac{4}{(d+2)T}\right)^{\frac{1}{d+4}} \xrightarrow{d=2} \hat{\sigma}\left(\frac{1}{T_{im}}\right)^{1/6}, \qquad \hat{\sigma} = \sqrt{\hat{\sigma}_x\,\hat{\sigma}_y}$$

over samples per image.

1. $\hat{h} = 26$ px is an arbitrary choice;
2. $\hat{h}_{im} = T_{im}^{-1/6}\,\hat{\sigma}_{im}$ for the ith participant and mth image;
3. $\hat{h}_M = \dfrac{1}{\sum_i M_i} \displaystyle\sum_i \sum_{m=1}^{M_i} \hat{h}_{im}$, the mean across images of all participants.

Tested for on-face duration weight.

4.2 Confirming basins are actually faces per the labelling

Scale down the set Ω_T similarly to G to get Ω_{st}. We test

$$O_i = \sum_{k=1}^{B_i} \mathbf{1}\Big\{ \min_v Q_v(\mu_k) \le 1 \Big\}; \qquad Q_v(\mu_k) = \left(\frac{x_k}{q_x}\right)^2 + \left(\frac{y_k}{q_y}\right)^2 \quad \text{over } \mu_k \in G,$$

where $v \mapsto (q_x, q_y)$ is the index specifying some face in an image, and B_i is the basins in an image i.

Sample a mean over the corner-shaved grid N times (chosen so that it is stable across resamples):

$$N = 100, \qquad U_i = \frac{1}{N} \sum_{b=1}^{N} \sum_{k=1}^{B_i} \mathbf{1}\Big\{ \min_v Q_v\big(u_k^{(b)}\big) \le 1 \Big\}, \qquad u_k^{(b)} \sim \mathcal{U}(G \cap \Omega_{st}).$$

The ratio tells us how likely basins are to be randomly placed due to the area of the screen, relative to the ratio of the ellipses formed by the centre against actual face regions (up to error in labelling faces).

We also check

$$A_i = \sum_{k=1}^{B_i'} \mathbf{1}\Big\{ \min_v Q_v\big(\mu_k^{(i')}\big) \le 1 \Big\},$$

where i' is a different participant over the same images. i' can either be for ABM as a conserved set, or for ABM as another random chance baseline removing central bias from uniform sampling (CTL); call these A_{iC}, A_{iU}, for C for IAPS which is always the same faces, expected strongest signal, and U for ABM random comparison, since across participants no image is related in face placement. The result is over a 6×6 grid (permute ABM, permute IAPS, same null, same non-null), all being different choices. Some examples are

$$\frac{O}{A_C} : 1.04,\ \text{IQR } (0.95\text{–}1.13),\ \chi^2_{28} = 54.2,\ p = 0.002 \qquad \text{for IAPS across participants}$$

$$\frac{O}{A_U} : 2.92,\ \text{IQR } (2.33\text{–}3.55);\ \text{Wilcoxon } p = 6 \times 10^{-8} \qquad \text{for ABM self null}$$

Concretely, faces are hit about equally across participants in IAPS, and ABM variance tells us basins on a face from one image are highly unlikely to be the same from another image, which matches the ABM structure in terms of faces.

4.3 Methods for the result tests

Dwell model. Dwell effects are fitted as

$$\log \Delta_{\text{dwell}} = \beta_0 + \beta_F\, \mathbf{1}_{\text{face}} + \beta_r\, r + \beta_{Fr}\, \mathbf{1}_{\text{face}}\, r + \beta_\theta \log \theta + \dots$$

with valence, depth, log face area and time since stimulus onset added where stated. A coefficient β corresponds to a $100(e^\beta - 1)\%$ change in dwell. Each model is fitted within each participant-session and the 29 coefficients tested against zero (Wilcoxon signed-rank).

Attendance. The unit is the participant–face pair: every face detected in every image a participant viewed, whether or not a basin landed on it. A face is attended if at least one basin lies on it. Attendance is modelled by logistic regression on log face area, log number of faces in the image, log eccentricity from screen centre, and valence.

Detailed balance ($J = 0$). For each trial the basins are listed in visit order and consecutive pairs counted, $N_{kl} = \#\{k \to l\}$. The asymmetry

$$A = \frac{\sum_{k,l} |N_{kl} - N_{lk}|}{\sum_{k,l} \big(N_{kl} + N_{lk}\big)}$$

is summed over all trials and compared with its distribution when each trial's basin sequence is randomly permuted (300 permutations), which keeps the basins and sequence length and destroys only the order. $J = 0$ predicts A at the null.

Differences between faces. Within each image-session with at least two visited faces, the between-face share of variance of a basin quantity x (log dwell, $\log\theta$, q) is

$$\frac{\sum_f n_f (\bar{x}_f - \bar{x})^2}{\sum_i (x_i - \bar{x})^2},$$

summed over image-sessions and compared with a null that permutes which basin is assigned to which face within each image-session. Permuting at the basin rather than visit level is required because θ and q are constant within a basin.

Stability across bandwidths and participants. Basins from two runs are matched if their centres lie within one kernel width (26 px). Persistence is the number of bandwidths (1–3) that find a basin at a location. Agreement of partitions is the adjusted Rand index between the basin labellings of the same trial on a 10 ms grid. On IAPS, consensus locations are formed by grouping basins from all participant-sessions within one kernel width; their count is the number of sessions placing a basin there.

Diffusion estimate. Kurtosis is computed on increment magnitudes $\|\Delta X\|$ between adjacent retained samples; a Gaussian increment distribution gives 3.

5 Results

Estimation

All dwell effects are fitted on log dwell. Because visits are nested in basins, images and sessions, each effect is estimated separately within each of the 29 participant-sessions and the 29 coefficients are then tested against zero (Wilcoxon signed-rank). The session is the unit, so no visit-level dependence enters the p-values; "$k/29$" is the number of sessions sharing the sign of the mean. Visit-level estimates with standard errors clustered on image-session agree in every case.

Faces hold gaze longer

On first entry, dwell on faces is 23% longer than off-face, controlling for basin pull (27/29 sessions, $p = 4 \times 10^{-8}$). The effect replicates on IAPS alone, the fully crossed stimulus set (+25%, 26/29, $p = 3 \times 10^{-7}$), and grows to +30% (28/29) when visits shorter than 30 ms are excluded. Across the three bandwidths the coefficient ranges 0.19–0.24.

Dwell is graded by centring

Within a face, dwell falls monotonically with distance from the face centre, from 330 to 280 ms between $q = 0$ and $q = 1$ (+18% centre vs. edge; 22/29, $p = 0.002$). The centre is the landmark centroid, so "centre" means near the nose rather than the eyes.

Face size determines which faces are looked at

30% of faces present receive a basin: 77.5% in single-face images, 25.7% in multi-face images, and 64.8% for the largest face in a multi-face image. Face area predicts attendance in every session (per-session logit $\beta = 3.75$, 29/29 positive; clustered on face $t = 17.6$), with competing faces and eccentricity secondary.

Valence acts only on faces

Off-face, Positive and Negative do not differ (+1.3%, 14/29, $p = 0.40$). On faces, Positive dwell is 13% shorter than Negative (23/29, $p = 3 \times 10^{-4}$). Positive faces are about 35% larger; with face area controlled the difference is 9% (18/29, $p = 0.02$). On IAPS alone it is not significant per session ($p = 0.21$). After centring and scaling, per participant × valence dwell distributions share one shape (Anderson–Darling $p \geq 0.25$): valence shifts location only.

Re-entry: brief grazes off-face, genuine decline on faces

Pooled dwell falls 17% per re-entry (29/29), robust to excluding visits cut off at stimulus offset and to time since onset, and replicated on IAPS. The decline, however, is not a shortening of sustained looking everywhere:

	off-face		on-face	
re-entry	brief (< 30 ms)	sustained median	brief	sustained median
0	2.4%	244 ms	3.7%	328 ms
1	8.7%	264	6.7%	362
2	14.6%	280	7.1%	362
3	19.3%	270	11.7%	346
4	23.1%	274	14.2%	304
5	27.0%	268	18.0%	280
6	28.6%	268	17.2%	274

Off-face, sustained visits do not shorten; the pooled decline comes entirely from the growing share of brief incursions at a basin's edge. With visits under 30 ms excluded the decay vanishes (−0.5%, $p = 0.27$). On faces, sustained dwell rises over the first two returns and then declines: per session, the median of visits ≥ 30 ms is 18.5% lower at $r \geq 4$ than at $r = 1$–2 (25/29 sessions, $p = 2 \times 10^{-4}$), while off-face it does not change (−1.4%, 16/29, $p = 0.64$). Brief grazes accumulate more slowly on faces than off-face.

Basins land on labelled faces

Basin centres fall inside labelled face ellipses far more often than reference point sets placed on the same images and faces. On ABM, relative to the same participant's basins from a different image, $O/A_U = 2.92$ (95% CI 2.59–3.28; per session median 3.06, IQR 2.33–3.55; 24/24 above 1; Wilcoxon $p = 6 \times 10^{-8}$). Relative to uniform placement the ratio is 3.05 (IAPS) and 6.08 (ABM). On IAPS, where every participant sees the same faces, a participant's hits relative to other viewers of the same pictures have median 1.04 (IQR 0.95–1.13): face capture is largely shared, with a small individual component that exceeds counting noise ($\chi^2_{28} = 54.2$, $p = 0.002$) and is stable across sessions ($\rho = 0.74$, $p = 0.01$).

Faces within an image differ

Against a basin-level permutation, faces in the same image differ in dwell ($z = +3.30$) and pull ($z = +3.73$), but only in Positive images ($z = +4.6$ vs. -0.7 for Negative). Relative centring is more similar across faces than chance ($z = -2.28$).

Model assumptions

$J = 0$ holds: transition asymmetry 0.653 against a shuffled-order null of 0.658 ± 0.002 ($z = -2.58$). Pull does not predict dwell: $\theta \cdot \Delta_{\text{dwell}} = 0.028$ against a predicted $O(1)$, log–log slope -0.13 against -1. The curvature λ is nearly constant across basins (s.d. 0.19 in log), so $\theta = D\lambda$ largely restates σ^2. Increments are leptokurtic (median kurtosis 14), and σ^2 differs between the binocular and monocular cohorts (84 vs. 304 px^2/s); basin geometry, membership, visits and dwell are D-invariant and unaffected.

Stability of the partition

Session ordering is preserved across the three bandwidths: over the 29 sessions, Spearman ρ is 0.89–0.99 for basins per image, 0.97 for on-face rate, 0.63–0.91 for median dwell and 0.88–1.00 for θ, and 74–88% of basins coincide within one kernel width. The Silverman rule gives $\hat{h}_M = 35.8$ px; with $\hat{\sigma}$ taken from the median absolute deviation its median is 25.1 px (IQR 14.0–39.4), against the 26 px used. Basins found under all three bandwidths sit on faces 2.5× more often than those found under one (0.157 vs. 0.064), and partitions agree across bandwidths (adjusted Rand index 0.74–0.83). On IAPS, locations shared by 17 or more participant-sessions are on faces 50% of the time, against 7% for locations found by a single session.

Summary

Finding	Evidence
Faces hold gaze longer	+23–30%, 27–28/29 sessions, IAPS replicates
Graded by centring	+18%, 22/29
Face size drives attendance	29/29 sessions
Valence acts only on faces	off-face null; on-face N > P by 9% with size controlled, 18/29
Re-entry	off-face: more brief grazes, no shortening; on-face: plateau then 18.5% decline, 25/29
Basins land on faces	2.9× same-person baseline (ABM), 24/24
Detailed balance ($J = 0$)	holds
Pull predicts dwell	no; θ restates σ^2
Partition stability	74–88% basin correspondence across h; consensus basins sit on faces

The model does what it was built for: it segments gaze trajectories into data-driven regions that correspond to labelled faces, stably across bandwidth and participants. Effects measured on that partition hold at the session level; its dynamical layer (θ, pull) does not describe residence time.